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GROUP 7 (K2)

Cloud-Native Learning Management System (LMS)

(Cloud-Native / Layered Architecture)

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1.0 System Overview

The Cloud-Native Learning Management System (LMS) is a web-based application designed to support online learning activities such as user authentication and course access. The system is developed using a layered architecture approach and deployed on cloud infrastructure, with Aiven.io providing a managed MySQL database service.

The system enables three main types of users, which are learners, educators, and administrators, to interact with the LMS through a web interface. Students can register, log in, and access course information, while educators and administrators manage courses and monitor system usage.

The architecture of the LMS is divided into three main layers. The Presentation Layer provides the user interface, allowing users to perform actions such as login, registration, and accessing the course dashboard. The Application Layer handles the core business logic, including user authentication, session management, and course retrieval based on user roles. The Data Access Layer is responsible for managing communication with the cloud database, performing operations such as storing user credentials, retrieving course data, and maintaining session records.

The system uses Aiven.io, a cloud-managed database service, to ensure high availability, scalability, and data reliability. By leveraging Aiven's distributed cloud infrastructure, the LMS can handle multiple user requests efficiently while maintaining data consistency and security.

Overall, the Cloud-Native LMS provides a scalable, secure, and maintainable platform for delivering educational services, aligning with modern cloud computing and software architecture principles.

2.0 Functional Requirements

The learning management system prototype focuses on two core modules that demonstrate the full architectural flow from the user interface to the Aiven.io cloud database.

Module 1: User Authentication

- FR-01: User Registration

The system shall allow new users to register by providing a username, email and password, which are then securely stored in the Aiven.io, MYSQL database.

- FR-02: Secure Login Authentication

The system shall verify user credentials against the stored records in the cloud database to grant access to the system.

- FR-03: Session Initialization

Upon successful login, the system shall create a session object to store user identity and maintain the login state during navigation.

- FR-04: Secure Logout

The system shall provide a logout function that terminates the user's session and redirects the user back to the login page.

Module 2: Course Dashboard

- FR-05: Authenticated Dashboard Access

The system shall ensure that only users with an active session can access the Course Dashboard module.

- FR-06: Dynamic Course Retrieval

The system shall fetch and display a list of available courses from the Aiven.io database based on the user's profile or enrollment status.

- FR-07: Course Detail View

The system shall allow users to select a specific course to view detailed information, such as course objectives and descriptions.

- FR-08: Session-Data Integration

The system shall display personalized information on the dashboard by retrieving data from the active session created in Module 1.

3.0 Non-Functional Requirements

- **Scalability:** The system must support independent scaling for each tier, allowing the Business Layer to handle increased traffic during peak periods, such as assignment submissions.
- **Availability:** By utilizing Aiven.io, the system aims for high availability through automated self-healing capabilities and distributed cloud nodes to minimize downtime.
- **Reliability:** The Data Access Layer must ensure data integrity and consistency through automated backups and secure SQL transactions within the Aiven cloud environment.
- **Security:** Access control must be strictly enforced, ensuring that only authenticated users with active sessions can access the dashboard and sensitive course materials.
- **Portability:** The architecture shall remain cloud-independent, allowing the LMS to be deployed or migrated across different cloud service providers without significant changes to the core logic.

4.0 Architecture Style Explanation

The Cloud-Native Learning Management System (LMS) is a web-based application developed using a layered architecture approach and deployed on cloud infrastructure. The system architecture is divided into three primary layers to ensure a separation of concerns and efficient delivery of educational services. The Presentation Layer serves as the user interface, providing web-based forms for user authentication and a dynamic course dashboard for learners, educators and administrators. Beneath this, the Application Layer handles core logic, including credential verification, session management and course retrieval based on user roles. The final layer, the Data Access Layer, manages communication with the cloud database using a Data Access Object (DAO) and a Cloud Connectivity Provider to perform secure operations such as storing credentials and maintaining session records.

By integrating these layers with cloud-native principles, the system leverages an Aiven.io managed MySQL service to ensure high availability, scalability and data reliability. This architectural style allows the Business Layer to scale independently to handle traffic spikes, while Aiven's distributed cloud infrastructure ensures multiple user requests are handled with consistent data security. Additionally, the architecture is designed for portability, remaining cloud-independent so the LMS can be migrated across different providers without requiring significant changes to the core logic. Overall, this layered, cloud-native approach provides a secure, scalable and maintainable platform that aligns with modern software architecture principles.

5.0 Service or Component Identification

1. Presentation Tier Components

- **User Authentication Interface:** A web-based form that captures user credentials (email/password) for registration and login processes.
- **Course Dashboard UI:** A dynamic interface that displays personalized course lists, descriptions and user profile information after a successful session is established.

2. Business Tier Services

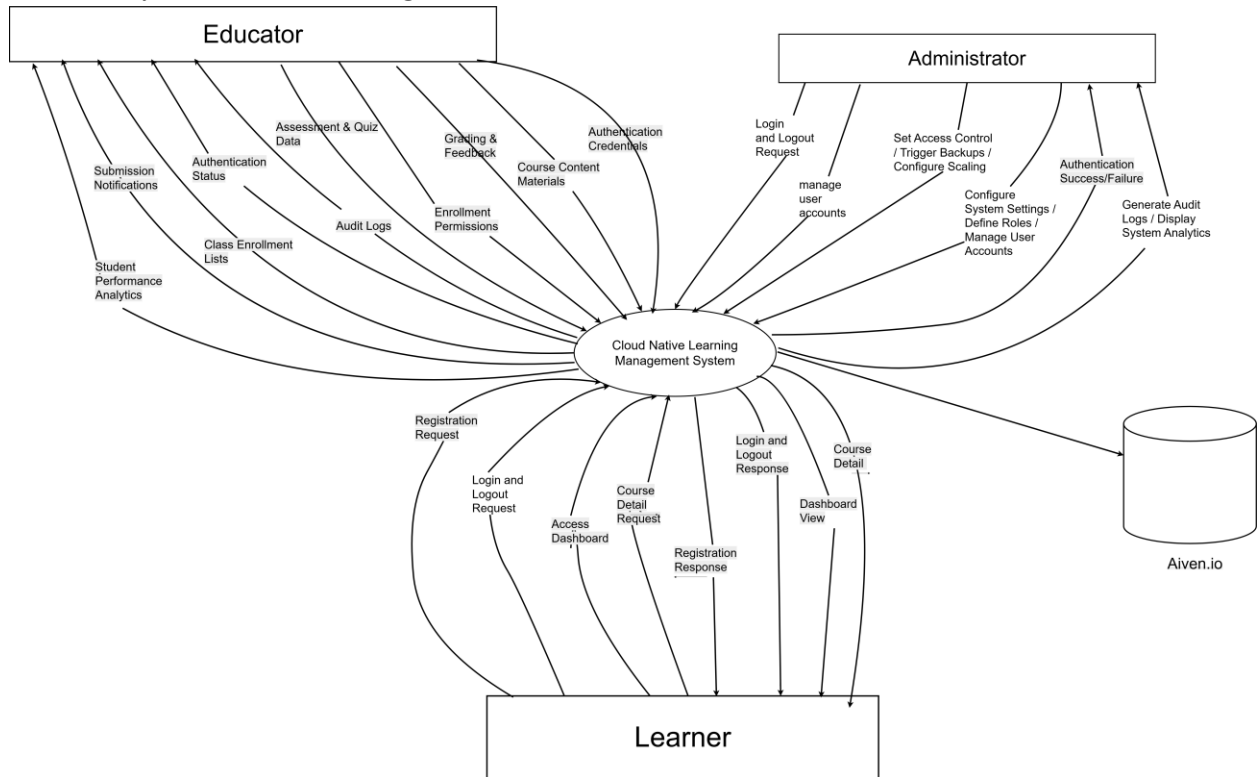
- **Authentication Service (Authentication Manager):** Responsible for validating user input, verifying credentials against the database and managing secure session objects.
- **Course Management Service:** Manages the logic for retrieving available courses from the cloud database based on the specific user's enrollment status.
- **Session Controller:** A security component that monitors active sessions to ensure only authenticated users can access restricted modules like the Course Dashboard.

3. Data Tier and Infrastructure Services

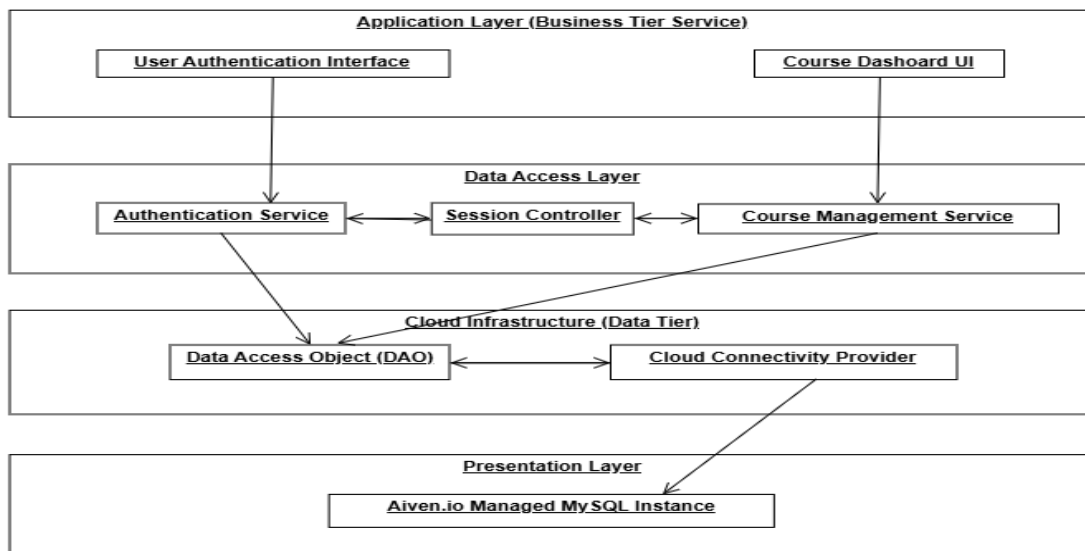
- **Data Access Object (DAO):** A structural component that provides an interface for the Business Tier to execute SQL queries (CRUD operations) without directly interacting with the database engine.
- **Aiven.io Managed MySQL Instance:** A cloud-native database service that provides high availability, automated backups and self-healing storage for all LMS data.
- **Cloud Connectivity Provider:** Ensures a secure connection between the application server and the Aiven.io cloud environment.

6.0 Architecture Diagrams

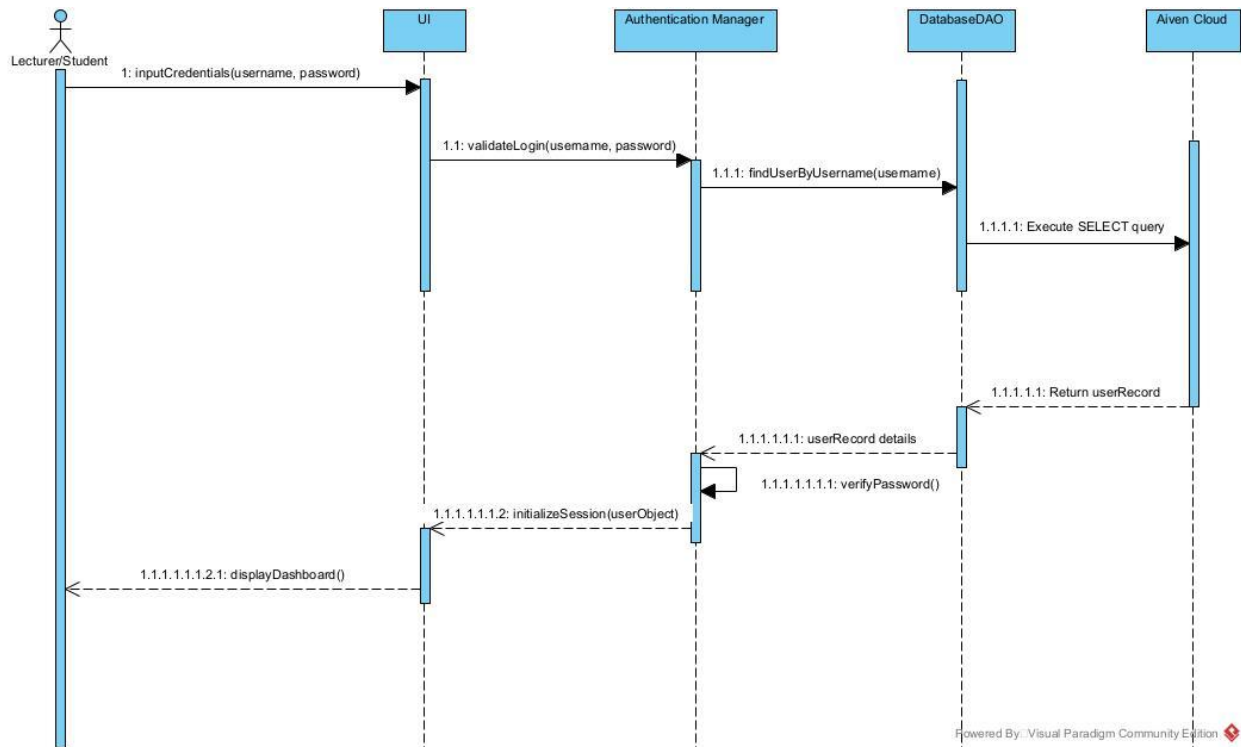
6.1 System Context Diagram



6.2 Component/Service Architecture Diagram



6.3 Interaction Diagram



6.4 Deployment Diagram

